

JevRL: Calibrated Typed Decisions from Language Models via Reward Learning

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Abstract. Language models are often asked to produce a single answer even when a downstream system needs a probability, a score, or an explicit abstention signal. We introduce JevRL, a minimal reward-learning interface that turns a frozen small language model into three typed outputs: a categorical choice, a scalar score, and a null/abstention probability. The central training primitive samples two independent answers and rewards agreement with a latent target while penalizing duplicate answers. Its expected reward is a quadratic proper objective, so the learner is encouraged to represent uncertainty rather than only maximize top-1 accuracy. We implement JevRL with a frozen Qwen2.5-0.5B-Instruct backbone and a 142K-parameter typed head. On a synthetic calibration benchmark with 600 training and 300 test items, JevRL reaches 0.611 accuracy, 0.093 true-target Brier score, and 0.040 expected calibration error. The one-seed pilot is deliberately small: it establishes the concept, compares supervised and reward baselines, and exposes the interface for larger language-model and real-task studies.

Contribution	Evidence
A typed interface for Choice, Score, and Noul.	Frozen Qwen2.5-0.5B + compact head.
A two-sample proper reward for uncertainty.	JevRL Brier 0.093 vs single-sample RL 0.372.
A runnable, container-first pilot protocol.	Slurm script, metrics, and demo included.

1. Introduction

A language model can be fluent while still being a poor interface for software. A free-form answer hides whether the model is choosing among candidates, estimating a continuous quantity, or deciding that it should abstain. The JevRL design treats these as typed decisions. A downstream program can consume a categorical distribution, a bounded score, and a null probability without parsing prose.

The goal of this report is intentionally modest. We make the concept concrete with a small, reproducible implementation and a synthetic benchmark that can run on one GPU. We do not claim a new large-scale reinforcement-learning algorithm, and we do not present a reproduction of any external Jev system. Instead, we ask a narrow question: can a frozen small language model support a reward-trained head whose outputs remain probabilistic and typed?

2. Problem formulation

Each item is a state x , a candidate set $C = \{c_1, \dots, c_K\}$, and a latent target y . The model encodes x and C into a representation h . JevRL exposes a distribution $p_{\theta}(c | x)$, a scalar $s_{\theta}(x)$ in $[0, 1]$, and a null probability $n_{\theta}(x)$. The first output answers *which candidate?*, the second answers *how much?*, and the third answers *should I abstain?*

For the reward-learning pilot, two independent categorical samples A and B are drawn from p_{θ} . The environment returns:

$$R(A, B, y) = 1[A = y] + 1[B = y] - 1[A = B].$$

If q is the one-hot target distribution and p is the model distribution, the expected reward is $\|q\|^2 - \|p - q\|^2$. Up to an additive constant, maximizing this reward is equivalent to minimizing a squared probability error. The duplicate penalty is

what prevents the two-sample estimator from rewarding a high-probability but overconfident answer alone.

3. Method

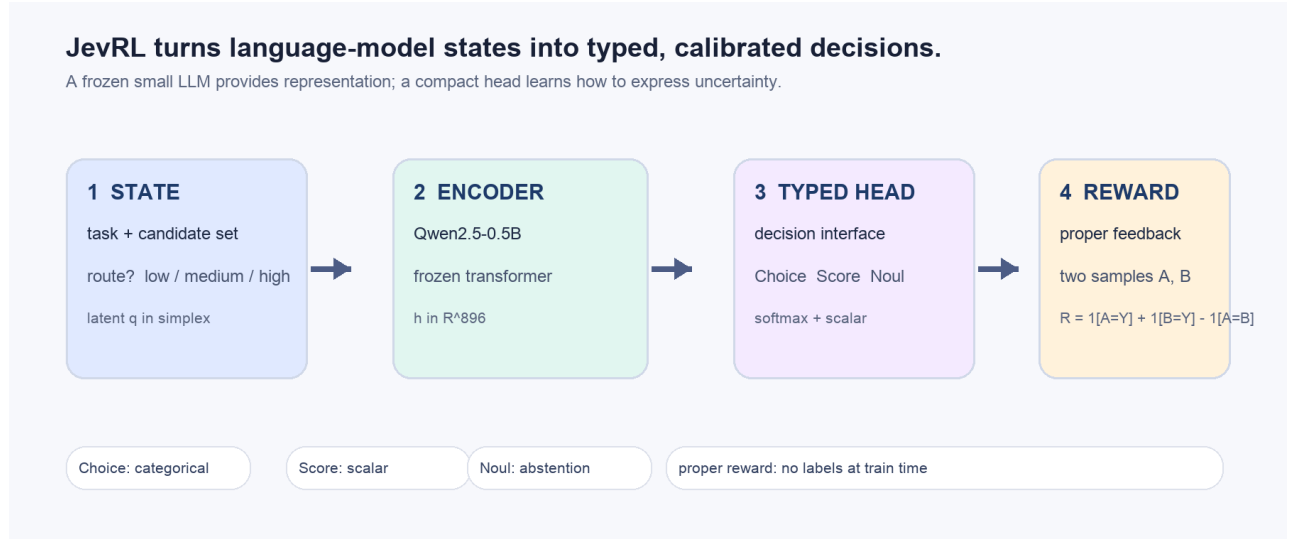


Figure 1: JevRL maps a state to a frozen language-model representation, then routes it through a small typed head. The reward is computed from two samples; the head remains a normal differentiable module for the supervised baselines.

3.1 Frozen backbone and typed head

We use Qwen2.5-0.5B-Instruct as a frozen text encoder. A masked mean of the final hidden states produces h in $\mathbb{R}^{896}$. A two-layer multilayer perceptron with 141,793 trainable parameters maps h to logits for Choice, a sigmoid Score, and a sigmoid Noul output. The backbone is never updated in the pilot, which keeps the experiment cheap and makes the effect of the typed interface easy to inspect.

3.2 Reward estimators

The JevRL objective uses the two-sample reward above. We compare it with a single-sample accuracy reward, which optimizes only whether one sampled answer is correct. We also include cross-entropy and Brier supervised heads that observe the target during training. For all heads, the hidden target probability q is withheld from training and used only for evaluation; this avoids leaking an oracle uncertainty signal.

3.3 Typed outputs

The three primitives are designed to be consumed independently. Choice is a categorical distribution over the candidate set, Score is a bounded scalar, and Noul is a null/abstention probability. In a production system, Noul could gate an escalation path; in this concept demo it is visualized and logged but not separately reward-optimized.

4. Benchmark and experimental setup

CalibTicket-Synth contains short support-ticket states generated from a fixed set of templates. Each item has three candidate actions and a latent target distribution q . We use 600 training items and 300 test items with seed 7. The benchmark is synthetic by design: it makes target uncertainty available for measuring calibration while avoiding any claim about a real customer-support workload.

Experiments run in a containerized Slurm job on one A2 H800 GPU with four CPUs and a 24 GB memory request. The model snapshot is Qwen2.5-0.5B-Instruct. We report top-1 accuracy, true-target Brier score, true-target negative log likelihood, and expected calibration error (ECE). Lower is better for the last three metrics.

5. Results

Pilot results on CalibTicket-Synth (one seed, 300 test items)

JevRL improves calibration over accuracy-only RL while matching the majority baseline on accuracy.

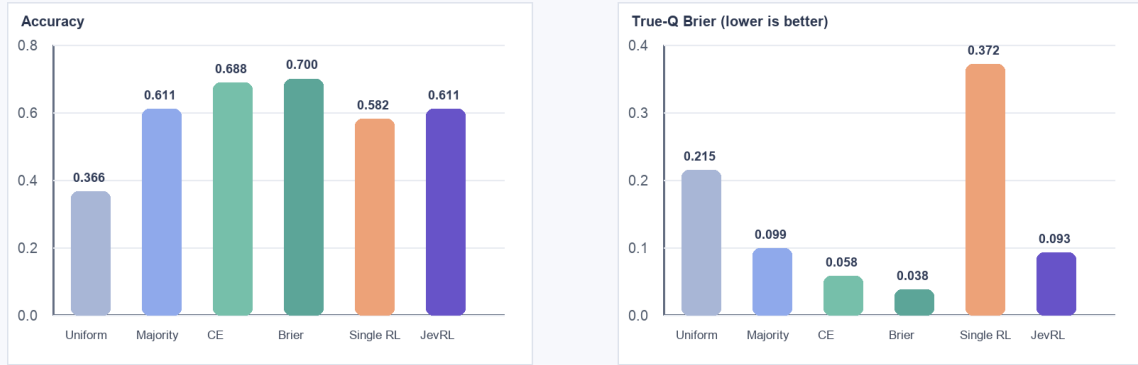


Figure 2: Pilot accuracy and calibration. JevRL matches the majority baseline in accuracy while avoiding the severe overconfidence of single-sample accuracy RL.

Method	Acc.	Brier ↓	NLL ↓	ECE ↓
Uniform	0.366	0.215	0.963	0.143
Majority	0.611	0.099	0.804	0.024
CE	0.688	0.058	0.754	0.054
Brier	0.700	0.038	0.727	0.060
Single-sample RL	0.582	0.372	2.199	0.393
JevRL (two-sample)	0.611	0.093	0.800	0.040

Table 1: One-seed pilot results. “True-target” metrics use q only for evaluation. The supervised heads see the hard target y during training; JevRL and single-sample RL use sampled rewards.

The pilot shows the intended tradeoff. Cross-entropy and Brier supervision obtain the highest accuracy, because they are directly optimized for the target. JevRL is less accurate than those supervised heads but substantially better calibrated than single-sample accuracy RL. Its Brier score is close to the majority baseline and its ECE is low, suggesting that the duplicate penalty preserves a useful probability signal.

A qualitative example is shown below. The typed interface keeps a categorical answer, a score, and an abstention signal separate even when they are computed from the same state.

One state, three typed outputs

The same latent state can feed different consumers without flattening everything into a free-form answer.

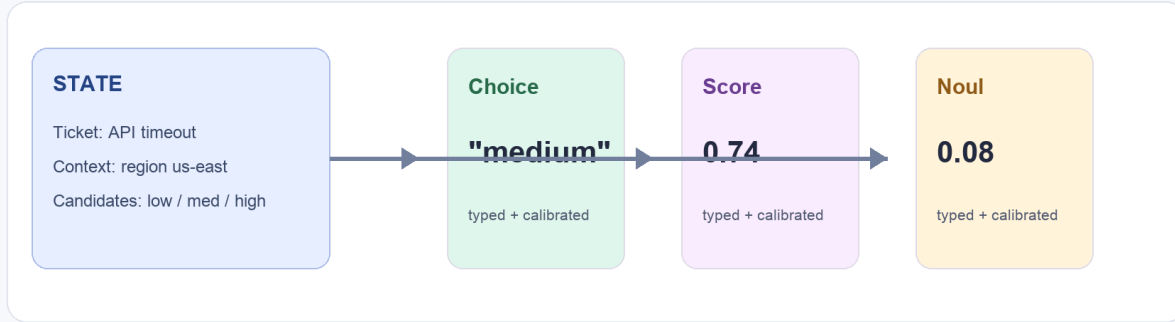


Figure 3: Example typed output for an API-timeout state. The values are a static visualization of the released demo interface.

6. Limitations and future work

This result has clear limits. The benchmark is synthetic, the run uses one random seed, and the trainable component is only a small head. We do not evaluate instruction following, long-context behavior, or robustness to distribution shift. The Noul output is exposed as an interface primitive but is not yet trained with a separate abstention dataset. Finally, the reward identity is a proper quadratic objective in the categorical setting; extending the same guarantee to structured, sequential, or tool-using actions requires additional analysis.

The next experiments should use a real decision benchmark with soft labels, repeat across seeds, and compare frozen and partially tuned backbones. A stronger study would train Noul on selective-prediction risk, test candidate-set permutations at scale, and measure whether typed outputs improve downstream tool routing. These are the steps needed before making claims about practical language-model agents.

7. Conclusion

JevRL is a small but concrete interface for reward-trained language-model decisions. A frozen Qwen2.5-0.5B backbone plus a compact typed head can produce a choice, a score, and an abstention probability. On a controlled pilot, two-sample proper feedback is markedly better calibrated than accuracy-only reward learning. The released code and page are intended as a reproducible starting point for larger, real-task studies.

References

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Reproducibility note. The released repository contains the toy implementation, Qwen-backed training script, Slurm launch file, pilot metrics, and the JSON used by the static demo. The paper reports the exact one-seed numbers available at release time.

Citation

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